A Companion to Serre's A Course in Arithmetic

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September 2026

Preface

I began writing this companion as a high school student working through Serre's *A Course in Arithmetic*. Serre is famously concise, so as I read, I often encountered skipped steps whose logic was unclear to me or mathematical terms I was unfamiliar with. Yet after filling in the missing background, the mathematics itself often revealed itself to be strikingly simple and even beautiful. I wanted to record some of my process in reading Serre, and I hope this might allow another reader to appreciate the beauty of Serre's book as well.

I don't intend this text to replace Serre, but rather to make a resource that can be read alongside it or consulted when needed. Problems are included throughout, inviting the reader to pause and work out an important step for themselves. Starred bonus problems are optional challenges, often arising from my own research or rabbit holes I've gone down. Their solutions are generally left as an exercise.

This guide is still unfinished: I am writing and revising it while currently reading Chapters I–IV. The eventual goal is to write a full accompaniment to these chapters, which culminate in the Hasse–Minkowski theorem and its proof.

Introduction

This is a companion to Chapters I-IV of Serre's *A Course in Arithmetic*, which culminate in the proof of the Hasse-Minkowski theorem.

Serre's statement of it is the following:

> **Theorem** (Hasse-Minkowski). In order that f represent 0, it is necessary and sufficient that, for all $v \in V$, the form f_v represent 0.

Simply making sense of this theorem statement requires keeping track of several definitions and notations introduced earlier in the book. What is f? What does it mean for f to *represent* 0? What is V? What is f_v? And so on.

By the end of Chapter IV, my goal is for neither this statement nor its proof to look cryptic.

In general, I will assume the reader is familiar with the following:

- Basic strategies of proof: direct, contradiction, induction, etc.
- Basic set and function notation: subsets, unions/intersections, functions (injective/surjective/bijective, composition, inverse, etc.)
- Basic polynomial algebra.
- Basic combinatorics/algebraic identities: binomial coefficients, binomial theorem, summation and product notation

- Elementary number theory: modular arithmetic, and the statement of Fermat's Little Theorem
- Basic linear algebra: vectors, matrices, dot products, bases, linear independence, dimension, determinants

Before beginning Chapter I, we will unpack these questions about the Hasse-Minkowski Theorem, using it as a small preview of the future format.

In particular, when Serre gives a definition, we will not only ask what it says, but also why each of its hypotheses is introduced. To do so, it's often useful to investigate what breaks if we remove one.

What is f?

Serre initially writes f as a quadratic form with rational coefficients:

$$f(X) = \sum_{i=1}^{n} a_{ii} X_i^2 + 2 \sum_{i<j} a_{ij} X_i X_j.$$

He also assumes that f is nondegenerate.

Serre later shows that, using a reversible change of variables, one can diagonalize the quadratic form, so that f can be written without the cross terms, as

$$f = a_1 X_1^2 + \cdots + a_n X_n^2.$$

Thus, in simple terms, nondegenerate means that after f is diagonalized, none of the a_i are 0. This excludes an easy case: if a quadratic form is degenerate, then one can easily construct a nontrivial zero. For instance, if $a_n = 0$, then

$$(0, \dots, 0, 1)$$

is already a nontrivial zero of f.

In this preview, we will work over the real numbers. Thus, we may think of a quadratic form as a function

$$Q : \mathbb{R}^n \to \mathbb{R}.$$

At first, one might try to define a quadratic form simply as a polynomial in which every term has total degree 2. However, Serre uses a more general and abstract definition that does not depend on a particular choice of coordinates. In our setting, we want a definition that characterizes exactly the functions which can be written in the form

$$\sum_{i=1}^{n} a_{ii} X_i^2 + 2 \sum_{i<j} a_{ij} X_i X_j.$$

In other words, every function of this form should satisfy our definition, and every function satisfying our definition should have this form.

In our setting, it becomes the following.

Definition (Quadratic Form) — A function

$$Q : \mathbb{R}^n \to \mathbb{R}$$

is called a *quadratic form* if

1. $Q(ax) = a^2 Q(x)$ for every $a \in \mathbb{R}$ and $x \in \mathbb{R}^n$, and
2. the function
$$(x, y) \mapsto Q(x + y) - Q(x) - Q(y)$$
is bilinear.

At first, this definition may seem much stranger than simply saying "a polynomial in which every term has degree 2." So before going any further, we should unpack the meaning of these two conditions.

Notably, quadratic forms are being viewed not as polynomials, but as functions from $\mathbb{R}^n$ to $\mathbb{R}$.

The first condition is already visible in an ordinary quadratic expression. If

$$Q(x, y) = x^2 - 2xy + 3y^2,$$

then

$$Q(ax, ay) = (ax)^2 - 2(ax)(ay) + 3(ay)^2 = a^2 Q(x, y).$$

That is, multiplying the input vector (x, y) by a multiplies the output by a^2, hence the term "quadratic."

Remark. Although in elementary algebra, expressions such as

$$h(x) = x^2 + 2x + 1$$

are considered quadratic polynomials, notice that

$$h(ax) = a^2x^2 + 2ax + 1 \neq a^2 h(x),$$

so by this definition, $h(x)$ is not a quadratic form. This is because terms such as $2x$ and 1 do not have degree 2.

So for an ordinary polynomial, the first condition forces every term to have degree 2.

But is this enough? Actually, it turns out to be too weak to be equivalent to our desired form. For instance, consider the piecewise function

$$f(x, y) = \begin{cases} x^2 + y^2, & xy \geq 0, \\ 2x^2 + 2y^2, & xy < 0. \end{cases}$$

One can verify that this function still satisfies

$$f(ax, ay) = a^2 f(x, y)$$

for every real number a. But this function is not of the form

$$a_{11}x^2 + 2a_{12}xy + a_{22}y^2.$$

To see why, suppose for contradiction that it were.

Problem (a) Evaluate the piecewise function f and the proposed quadratic form at $(1,0)$ and $(0,1)$. What does this tell you about a_{11} and a_{22}?

(b) Find $f(1,1)$, and from there solve for a_{12}.

(c) Find a contradiction.

Solution. *Since*

$$f(1,0) = 1, \qquad f(0,1) = 1,$$

we would have

$$a_{11} = a_{22} = 1.$$

Since

$$f(1,1) = 2,$$

we would then have

$$1 + 2a_{12} + 1 = 2,$$

so $a_{12} = 0$. *But this would force, for instance,*

$$f(1,-1) = 1 + 1 = 2,$$

whereas our definition gives

$$f(1,-1) = 4.$$

Thus no quadratic polynomial of our desired form can produce this function.

The first condition tells us how the value changes when we scale a vector, but it does not tell us how the values at two different vectors relate to the value at their sum. Hence, we need another condition that connects these values. Thus, we have motivated the second condition. Let us investigate the relationship between $Q(x+y)$ and $Q(x)+Q(y)$.

First, we consider the single-variable case. Let

$$Q(t) = t^2.$$

Then

$$Q(u+v) = (u+v)^2 = u^2 + 2uv + v^2.$$

Subtracting gives

$$Q(u+v) - Q(u) - Q(v) = 2uv.$$

Does something similar happen for quadratic forms in two variables? Suppose for fixed coefficients $a, b, c \in \mathbb{R}$,

$$Q(x_1, x_2) = ax_1^2 + 2bx_1x_2 + cx_2^2.$$

Take two input vectors $x = (x_1, x_2)$ and $y = (y_1, y_2)$. Their sum is $x + y = (x_1 + y_1, x_2 + y_2)$, so we obtain

$$\begin{aligned} Q(x+y) &= a(x_1+y_1)^2 + 2b(x_1+y_1)(x_2+y_2) + c(x_2+y_2)^2 \\ &= \underbrace{ax_1^2 + 2bx_1x_2 + cx_2^2}_{Q(x)} + \underbrace{ay_1^2 + 2by_1y_2 + cy_2^2}_{Q(y)} + 2ax_1y_1 + 2bx_1y_2 + 2by_1x_2 + 2cx_2y_2. \end{aligned}$$

Hence,

$$Q(x+y) - Q(x) - Q(y) = 2ax_1y_1 + 2bx_1y_2 + 2by_1x_2 + 2cx_2y_2.$$

Just as in the single-variable example, every remaining term contains one factor from each input vector. To formalize this, we say that the function $(x,y) \mapsto Q(x+y) - Q(x) - Q(y)$ is **bilinear.**

Definition (Linear) — A function $f : \mathbb{R}^n \to \mathbb{R}$ is **linear** if it preserves addition and scalar multiplication. That is, for all $u, v \in \mathbb{R}^n$ and $t \in \mathbb{R}$,

$$f(u+v) = f(u) + f(v), \qquad f(tu) = tf(u).$$

Remark. In elementary algebra, functions of the form

$$f(x) = mx + b$$

are often said to be linear. Under our definition, however, this function is linear only when $b = 0$. In fact, every linear function must satisfy

$$f(0) = 0.$$

To see this, set $t = 0$ in the identity $f(tu) = tf(u)$. We obtain

$$f(0) = f(0u) = 0f(u) = 0.$$

But for $f(x) = mx + b$, we have $f(0) = b$. Thus, if $b \neq 0$, the function is not linear in our sense.

Definition (Bilinear) — A function $B : \mathbb{R}^n \times \mathbb{R}^n \to \mathbb{R}$ is **bilinear** if it is linear in each input separately. That is, when y is held fixed, $B(x, y)$ is linear as a function of x, and when x is held fixed, $B(x, y)$ is linear as a function of y.

Formally, for all $u, v, w \in \mathbb{R}^n$ and $t \in \mathbb{R}$,

$$\begin{aligned} B(u+v, w) &= B(u, w) + B(v, w), \quad B(tu, w) = tB(u, w), \\ B(w, u+v) &= B(w, u) + B(w, v), \quad B(w, tu) = tB(w, u). \end{aligned}$$

Returning to our example, let

$$B(x, y) = Q(x+y) - Q(x) - Q(y).$$

We found that

$$B(x, y) = 2ax_1y_1 + 2bx_1y_2 + 2by_1x_2 + 2cx_2y_2.$$

Problem

For fixed constants a, b, verify that the function

$$f : \mathbb{R}^2 \to \mathbb{R}, \qquad f(x) = ax_1 + bx_2$$

is linear.

Solution. *We see that*

$$f(x+y) = a(x_1+y_1) + b(x_2+y_2) = (ax_1+bx_2) + (ay_1+by_2) = f(x) + f(y).$$

Also,

$$f(tx) = atx_1 + btx_2 = t(ax_1+bx_2) = tf(x),$$

so f is linear.

Problem

Show that B is bilinear by grouping terms to check linearity in each input separately.

Solution. *Grouping by the coordinates of x gives*

$$B(x,y) = (2ay_1 + 2by_2)x_1 + (2by_1 + 2cy_2)x_2.$$

When y is fixed, the coefficients in parentheses are constants, so this expression is linear in x by the previous problem. Similarly, grouping by the coordinates of y gives

$$B(x,y) = (2ax_1 + 2bx_2)y_1 + (2bx_1 + 2cx_2)y_2,$$

which is similarly linear in y when x is fixed. Hence B is bilinear.

We have now seen why quadratic expressions of the desired form satisfy Serre's two conditions. But do those conditions rule out every other kind of function? The following problem shows that they do.

Problem

Suppose $Q : \mathbb{R}^n \to \mathbb{R}$ satisfies the two conditions in the definition of a quadratic form. Define

$$B(x,y) = Q(x+y) - Q(x) - Q(y),$$

and let $e_1, \dots, e_n$ be the standard basis vectors of $\mathbb{R}^n$.

(a) Use the first condition to show that

$$B(x,x) = 2Q(x).$$

(b) Show directly from definition that

$$B(x,y) = B(y,x).$$

(c) Recall we can write a vector $x = (x_1, x_2)$ as $x = x_1e_1 + x_2e_2$. In general, write $x = x_1e_1 + \cdots + x_ne_n$. Using bilinearity, expand $B(x,x)$ in terms of the values $B(e_i, e_j)$.

(d) Combine the terms corresponding to (i,j) and (j,i) to show that there exist real numbers a_{ij} such that

$$Q(x) = \sum_{i=1}^{n} a_{ii}x_i^2 + 2\sum_{i<j} a_{ij}x_ix_j.$$

Solution. *(a) Using the first condition, $Q(ax) = a^2Q(x)$, we have*

$$B(x,x) = Q(2x) - Q(x) - Q(x) = 4Q(x) - 2Q(x) = 2Q(x).$$

(b) We have

$$\begin{aligned} B(x,y) &= Q(x+y) - Q(x) - Q(y) \\ &= Q(y+x) - Q(y) - Q(x) \\ &= B(y,x). \end{aligned}$$

(c) We have $x = x_1 e_1 + \cdots + x_n e_n = \sum_{i=1}^{n} x_i e_i$. So, by bilinearity,

$$\begin{aligned} B(x,x) &= B\left(\sum_{i=1}^{n} x_i e_i, \sum_{j=1}^{n} x_j e_j\right) \\ &= \sum_{i=1}^{n}\left[x_i\, B\left(e_i, \sum_{j=1}^{n} x_j e_j\right)\right] \\ &= \sum_{i=1}^{n}\left[x_i \sum_{j=1}^{n}\big(x_j\, B\left(e_i, e_j\right)\big)\right] \\ &= \sum_{i=1}^{n}\sum_{j=1}^{n} x_i x_j\, B\left(e_i, e_j\right). \end{aligned}$$

(d) As before,

$$Q(x) = \frac{1}{2}B(x,x) = \frac{1}{2}\sum_{i=1}^{n}\sum_{j=1}^{n} x_i x_j B(e_i, e_j).$$

Now, we separate the terms where $i = j$ from those where $i \neq j$. Pairing the term corresponding to (i,j) with the term corresponding to (j,i), we obtain

$$\begin{aligned} Q(x) &= \frac{1}{2}\sum_{i=1}^{n} x_i^2 B(e_i, e_i) + \frac{1}{2}\sum_{i<j} x_i x_j \big(B(e_i, e_j) + B(e_j, e_i)\big) \\ &= \frac{1}{2}\sum_{i=1}^{n} x_i^2 B(e_i, e_i) + \sum_{i<j} x_i x_j B(e_i, e_j), \end{aligned}$$

For every i and j, define

$$a_{ij} = \frac{1}{2}B(e_i, e_j).$$

Since B takes values in $\mathbb{R}$, each a_{ij} is a real number. Substituting this definition into the preceding expression gives

$$Q(x) = \sum_{i=1}^{n} a_{ii} x_i^2 + 2\sum_{i<j} a_{ij} x_i x_j,$$

as desired.

So far, we've checked both directions: quadratic expressions of the desired form satisfy Serre's two conditions, and most recently, the converse: every function satisfying those conditions must have the desired form.

In fact, we haven't just proven that the coefficients a_{ij} exist, but also shown how to derive them from Q. That is,

$$a_{ij} = \frac{1}{2}B(e_i, e_j) = \frac{1}{2}\big[Q(e_i + e_j) - Q(e_i) - Q(e_j)\big]$$

These kinds of constructive proofs, rather than simply proofs of existence, are especially satisfying kinds of proof.

The below example illustrates why this result is useful. We begin with a function defined based on distances, without choosing a coordinate system. Once we verify Serre's two conditions, we can represent the function by a quadratic expression.

Problem

Let u, v be vectors in the plane such that

$$\|u\| = 2, \qquad \|v\| = 3, \qquad \|u - v\| = \sqrt{7}.$$

Define $Q(s,t)$ to be the squared distance from the origin to the point $su + tv$, that is,

$$Q(s,t) = \|su + tv\|^2.$$

Find an explicit formula for Q. You might find the following identities useful, where $\cdot$ denotes the dot product:

- For vector v, $\|v\|^2 = v \cdot v$.
- For vector v and scalar a, $\|av\|^2 = a^2\|v\|^2$.
- *Polarization identity:* For vectors z, w, $\|z + w\|^2 - \|z\|^2 - \|w\|^2 = 2(z \cdot w)$.
- For vectors v, w, the dot product $v \cdot w$ is bilinear (it distributes over vector addition and respects scalar multiplication for each input).

Solution. *First we verify that $Q(as, at) = a^2 Q(s,t)$. We have*

$$\begin{aligned} Q(as, at) &= \|asu + atv\|^2 \\ &= a^2\|su + tv\|^2 \\ &= a^2 Q(s,t). \end{aligned}$$

Now take two inputs

$$x = (s_1, t_1), \qquad y = (s_2, t_2),$$

and set

$$z = s_1 u + t_1 v, \qquad w = s_2 u + t_2 v.$$

Then the sum $x + y$ is mapped to the vector $z + w$. Hence, by the polarization identity,

$$\begin{aligned} Q(x + y) - Q(x) - Q(y) &= \|z + w\|^2 - \|z\|^2 - \|w\|^2 \\ &= 2(z \cdot w). \end{aligned}$$

Since z, w depend linearly on x, y respectively, and the dot product is bilinear, we know $Q(x + y) - Q(x) - Q(y)$ is bilinear.

Hence, let $B(x, y) = Q(x + y) - Q(x) - Q(y)$. We calculate

$$Q(1, 0) = \|u\|^2 = 4, \qquad Q(0, 1) = \|v\|^2 = 9.$$

We also need to find $u \cdot v$:

$$\begin{aligned} \|u - v\|^2 &= \|u\|^2 + \|v\|^2 - 2(u \cdot v) \\ 7 &= 13 - 2(u \cdot v) \\ u \cdot v &= 3. \end{aligned}$$

Hence,

$$Q(1, 1) = \|u + v\|^2 = \|u\|^2 + \|v\|^2 + 2(u \cdot v) = 4 + 9 + 2(3) = 19,$$

and so, by bilinearity,

$$\begin{aligned}
a_{11} &= \frac{1}{2}B(e_1, e_1) = \frac{1}{2}(Q(2,0) - 2Q(1,0)) = Q(1,0) = 4. \\
a_{12} &= \frac{1}{2}B(e_1, e_2) = \frac{1}{2}(19 - 4 - 9) = 3 \\
a_{22} &= \frac{1}{2}B(e_2, e_2) = \frac{1}{2}(Q(0,2) - 2Q(0,1)) = Q(0,1) = 9.
\end{aligned}$$

Hence our final expression for $Q(s,t)$ is

$$Q(s,t) = a_{11}s^2 + 2a_{12}st + a_{22}t^2 = 4s^2 + 6st + 9t^2.$$

Remark. Miraculously, the preceding solution didn't require us to choose coordinates for u or v in order to solve the problem! For comparison, here is a coordinate-based solution.

Without loss of generality, we can let $u = (2, 0)$. Then

$$\begin{aligned}
\|u - v\| &= \sqrt{(u_1 - v_1)^2 + (u_2 - v_2)^2} \\
&= \sqrt{(u_1^2 + u_2^2) + (v_1^2 + v_2^2) - (2u_1v_1 + 2u_2v_2)} \\
&= \sqrt{2^2 + 3^2 - 2(2v_1)} \\
&= \sqrt{13 - 4v_1}.
\end{aligned}$$

Hence

$$13 - 4v_1 = 7 \implies v_1 = 3/2.$$

We can take the positive value for v_2, for instance:

$$v_2 = \sqrt{9 - 9/4} = \frac{3\sqrt{3}}{2}.$$

Thus,

$$su + tv = \left(2s + \frac{3}{2}t, \frac{3\sqrt{3}}{2}t\right).$$

Finally,

$$\begin{aligned}
\|su + tv\|^2 &= \left(2s + \frac{3}{2}t\right)^2 + \frac{27}{4}t^2 \\
&= 4s^2 + 6st + \frac{9}{4}t^2 + \frac{27}{4}t^2 \\
&= 4s^2 + 6st + 9t^2.
\end{aligned}$$

Notice also that unlike this solution, the preceding solution did not require referring to the components of u and v individually.

What does it mean to represent 0?

Now, what does it mean for f to represent 0?

Definition — A form $f(X_1, \ldots, X_n)$ represents 0 over a number system K if there exists a nonzero vector $x \in K^n$ such that $f(x) = 0$.

In other words, f represents 0 if and only if there is some *nontrivial* solution to

$$f(x_1, \ldots, x_n) = 0.$$

What are $V, \mathbb{Q}_v$, and f_v?

Serre defines V as the set of prime integers along with ∞. Thus

$$V = \{2, 3, 5, 7, \ldots, \infty\}$$

To define f_v, we first have to define $\mathbb{Q}_v$.

First, v is an element of V. This means either v is a prime integer p, or $v = \infty$. In the latter case, we say $\mathbb{Q}_\infty = \mathbb{R}$. In the former case, $\mathbb{Q}_v = \mathbb{Q}_p$, which is the field of p-adic numbers, developed in Chapter II.

For now, we can think of $\mathbb{Q}_p$ as a number system which contains the rationals and is associated with the prime p.

In fact, every rational number can be regarded as a distinct element of $\mathbb{Q}_v$, with their addition and multiplication remaining intact. Hence, a polynomial equation satisfied by rational numbers remains true when those same numbers are viewed in $\mathbb{Q}_v$.

Thus we may view f, a quadratic form with rational coefficients, as a quadratic form over $\mathbb{Q}_v$. The form f does not change as written, but its coefficients are interpreted as coefficients in $\mathbb{Q}_v$, and its set of possible inputs is expanded from vectors in $\mathbb{Q}^n$ to vectors in $\mathbb{Q}_v^n$. For instance, $f_3(X, Y)$ allows X, Y to be 3-adic numbers, not just rationals.

We denote this same quadratic form, now interpreted over $\mathbb{Q}_v$, by f_v.

Final Explanation

We are finally ready to return to the Hasse–Minkowski theorem, and translate it.

Theorem (Hasse–Minkowski)**.** In order that f represent 0, it is necessary and sufficient that, for all $v \in V$, the form f_v represent 0.

Recall that V consists of every prime together with ∞. Thus, saying that f_v represents 0 for every $v \in V$ means that the equation

$$f(x_1, \ldots, x_n) = 0$$

has a nontrivial solution over $\mathbb{R}$ and also over $\mathbb{Q}_p$ for every prime p.

In other words, the Hasse–Minkowski theorem says,

A nondegenerate quadratic form with rational coefficients has a nontrivial rational zero if and only if it has a nontrivial real zero and a nontrivial p-adic zero for every prime p.

One direction follows immediately. Suppose f has a nontrivial rational zero

$$(x_1, \ldots, x_n) \in \mathbb{Q}^n.$$

Those same rational coordinates can be viewed in every $\mathbb{Q}_v$, without changing their arithmetic. Thus, the same vector gives a nontrivial zero of f_v for every $v \in V$. This is the necessary direction.

The converse is the remarkable direction: suppose f has a nontrivial zero over $\mathbb{R}$ and over $\mathbb{Q}_p$ for every prime p. These solutions may in fact be different in every number system: for example, the 2-adic solution need not equal the 3-adic solution, and neither needs to map directly back into $\mathbb{Q}$. Remarkably, Hasse–Minkowski theorem says that the separate existence of all these possibly distinct solutions forces a rational solution to exist.

A zero over $\mathbb{Q}$ is called *global*, while a zero over one of the fields $\mathbb{Q}_v$ is called *local*. Thus, Hasse–Minkowski is an example of Hasse's local-global principle: a quadratic form has a nontrivial global zero if and only if it has a nontrivial local zero everywhere.

This is the main result toward which Chapters I–IV are building.

General Notation

Here, $\mathbb{Z}$ denotes the set of integers:

$$\dots, -2, -1, 0, 1, 2, \dots,$$

while, $\mathbb{Q}$ denotes the set of rational numbers, and $\mathbb{R}$ the set of real numbers.

1 Chapter 1: Finite Fields

1.1 Generalities

1.1.1 Background and Notation

For a positive integer n, we write

$$\mathbb{Z}/n\mathbb{Z}$$

for the integers modulo n. To understand what this notation means, consider arithmetic modulo 5. We have

$$2 \equiv 7 \equiv 12 \equiv -3 \pmod 5,$$

because each of these integers leaves the same "remainder" when divided by 5. Equivalently, the difference between any two of them is divisible by 5.

Thus, we regard all of these integers as equivalent mod 5, and we call the set

$$\{\dots, -8, -3, 2, 7, 12, \dots\}$$

a **residue class** modulo 5.

Definition 1.1.1 (Residue Class) — The *residue class* of an integer a modulo n is the set of all integers congruent to a modulo n, that is,

$$\{x \in \mathbb{Z} : x \equiv a \pmod n\}$$

Problem 1.1.2

How many *distinct* residue classes are there modulo 5?

One can show there are exactly 5 such classes, which we may represent by $0, 1, 2, 3, 4$.

More generally, $\mathbb{Z}/n\mathbb{Z}$ is the set of residue classes modulo n. We can add and multiply these classes by choosing representatives and then reducing modulo n. For example, in $\mathbb{Z}/5\mathbb{Z}$,

$$2 + 4 \equiv 6 \equiv 1 \pmod 5$$

and

$$2 \cdot 4 \equiv 8 \equiv 3 \pmod 5.$$

When p is prime, we will also write

$$\mathbb{F}_p = \mathbb{Z}/p\mathbb{Z}.$$

Remark 1.1.3. Some texts use the symbol

$$\mathbb{Z}_p$$

informally to denote the integers modulo p. Serre does not use this notation in that way. Instead, in *A Course in Arithmetic*, $\mathbb{Z}_p$ denotes the **ring of p-adic integers**, which will be introduced in Chapter II. Thus we will always write

$$\mathbb{Z}/p\mathbb{Z} \qquad \text{or} \qquad \mathbb{F}_p$$

for the integers modulo p, and reserve $\mathbb{Z}_p$ for the p-adic integers.

Now, notice that both $\mathbb{Z}$ and $\mathbb{Z}/n\mathbb{Z}$ are not simply sets. They have two familiar operations associated with them: addition and multiplication. In fact, if we add or multiply any two elements, the result stays inside the same set.

These are two examples of a more general structure called a **ring.**

Definition 1.1.4 (Ring) — A *ring* is a set R equipped with two binary operations, called addition and multiplication, satisfying the following properties:

1. Associativity: For all $a, b, c \in R$,
$$(a+b)+c = a+(b+c), \qquad (ab)c = a(bc).$$
2. Commutativity: For all $a, b \in R$,
$$a+b = b+a.$$
3. Zero: There exists an additive identity $0 \in R$, such that for all $a \in R$,
$$a+0 = a.$$
4. Negatives: For all $a \in R$, there exists an additive inverse $-a \in R$ such that
$$a+(-a) = 0.$$
5. One: There exists a multiplicative identity $1 \in R$, such that for all $a \in R$,
$$a \cdot 1 = a.$$
6. Distributivity: For all $a, b, c \in R$,
$$a(b+c) = ab+ac, \qquad (a+b)c = ac+bc.$$

Notice that multiplication isn't required to be commutative. However, Serre will usually work with commutative rings, meaning that we additionally require

$$ab = ba$$

for all $a, b \in R$. Both $\mathbb{Z}$ and $\mathbb{Z}/n\mathbb{Z}$ have this property.

But rings such as $\mathbb{Z}$ still lack something that we do have in $\mathbb{Q}$ and $\mathbb{R}$. In $\mathbb{Q}$, every nonzero number has a **multiplicative inverse**, in other words, that division into 1 is defined. For instance, in $\mathbb{Q}$,

$$2 \cdot \frac{1}{2} = 1,$$

So $\frac{1}{2}$ is the multiplicative inverse of 2 in $\mathbb{Q}$. However, there is no integer z for which $2z = 1$; 2 does not have a multiplicative inverse in $\mathbb{Z}$. This distinction motivates us to introduce a key object of Chapter I: a *field*.

Fields. As seen above, although $\mathbb{Z}$ and $\mathbb{Q}$ are both rings, a key difference between them is that every nonzero element in $\mathbb{Q}$ has a multiplicative inverse, which is not true for $\mathbb{Z}$.

This suggests the following definition.

Definition 1.1.5 (Field) — A *field* is a commutative ring K in which every nonzero element has a multiplicative inverse. In other words, for every $a \in K$ with $a \neq 0$, there exists some $a^{-1} \in K$ such that

$$aa^{-1} = 1.$$

Note that a number can be its own multiplicative inverse. For instance, $(-1) \cdot (-1) = 1$.

The number systems $\mathbb{Q}$ and $\mathbb{R}$ are fields. The integers $\mathbb{Z}$ are not.

What about the rings $\mathbb{Z}/n\mathbb{Z}$ that we discussed above?

Problem 1.1.6

Determine whether $\mathbb{Z}/5\mathbb{Z}$ and $\mathbb{Z}/6\mathbb{Z}$ are fields. If not, which nonzero elements do not have a multiplicative inverse?

Solution. *Consider first $\mathbb{Z}/5\mathbb{Z}$. Its elements may be represented by*

$$0, 1, 2, 3, 4.$$

We see that

$$1 \cdot 1 \equiv 2 \cdot 3 \equiv 3 \cdot 2 \equiv 4 \cdot 4 \equiv 1 \pmod 5.$$

Thus every nonzero element of $\mathbb{Z}/5\mathbb{Z}$ has a multiplicative inverse. Therefore, $\mathbb{Z}/5\mathbb{Z}$ is a field.

Now, in $\mathbb{Z}/6\mathbb{Z}$, the element 2 has no multiplicative inverse modulo 6. Indeed,

$$2 \cdot 0 \equiv 0, \quad 2 \cdot 1 \equiv 2, \quad 2 \cdot 2 \equiv 4, \quad 2 \cdot 3 \equiv 0, \quad 2 \cdot 4 \equiv 2, \quad 2 \cdot 5 \equiv 4 \pmod 6,$$

so none of these products is congruent to 1. An alternate approach would be to assume for contradiction that there exists some $x \in \mathbb{Z}/6\mathbb{Z}$ such that $2x \equiv 1 \pmod 6$. Then for some $q \in \mathbb{Z}$,

$$2x = 1 + 6q,$$

so

$$2(x - 3q) = 1.$$

We know $x - 3q \in \mathbb{Z}$, but there is no such integer z such that $2z = 1$. Hence 2 has no multiplicative inverse in $\mathbb{Z}/6\mathbb{Z}$. Similar computations reveal that 3 and 4 do not have multiplicative inverses, but 1 and 5 do, as $1 \cdot 1 \equiv 5 \cdot 5 \equiv 1 \pmod 6$.

We conclude that $\mathbb{Z}/6\mathbb{Z}$ is not a field.

What is different about 5 and 6?

The key point is that 5 is prime, while 6 is not. In fact, we can show that if p is prime, every nonzero element of

$$\mathbb{Z}/p\mathbb{Z}$$

has a multiplicative inverse. One avenue of proof is through Fermat's Little Theorem. Another is through the optional bonus problem 1.1.12.

Proposition 1.1.1. If p is prime, then $\mathbb{Z}/p\mathbb{Z}$ is a field.

Proof. Let $a \neq 0$ in $\mathbb{Z}/p\mathbb{Z}$. By Fermat's Little Theorem,

$$a^{p-1} \equiv 1 \pmod p.$$

Therefore,

$$a(a^{p-2}) \equiv 1 \pmod p,$$

so every nonzero element of $\mathbb{Z}/p\mathbb{Z}$ has a multiplicative inverse. $\square$

This allows us to write

$$\mathbb{F}_p = \mathbb{Z}/p\mathbb{Z}.$$

This gives us our first examples of **finite fields**: fields containing only finitely many elements.

For instance,

$$\mathbb{F}_2 = \{0, 1\}, \qquad \mathbb{F}_3 = \{0, 1, 2\}, \qquad \mathbb{F}_5 = \{0, 1, 2, 3, 4\},$$

where all arithmetic is performed modulo the corresponding prime.

Serre begins by letting K be a field and considering the image of the integers inside K.

Every field has a multiplicative identity, which we call 1. Starting from this element, we can repeatedly add it to itself:

$$1, \qquad 1+1, \qquad 1+1+1, \qquad \ldots$$

This gives us a natural way to send positive integers into K. Namely,

$$1 \mapsto 1, \qquad 2 \mapsto 1+1, \qquad 3 \mapsto 1+1+1,$$

and so on.

We include the nonpositives as well, with

$$0 \mapsto 0,$$

and using additive inverses, with

$$-1 \mapsto -1, \qquad -2 \mapsto -(1+1),$$

and so on. Thus there is a map

$$\phi : \mathbb{Z} \to K$$

defined by

$$\phi(n) = n \cdot 1_K.$$

Here, $n \cdot 1_K$ means adding the multiplicative identity 1_K to itself n times when $n > 0$, extending the definition as above to 0 and negative integers.

Importantly, this map respects addition and multiplication:

$$\phi(m+n) = \phi(m) + \phi(n),$$

$$\phi(mn) = \phi(m)\phi(n).$$

So arithmetic in $\mathbb{Z}$ is also carried into arithmetic in K. But do different integers always remain different after we send them into K?

In some fields such as $\mathbb{Q}$ or $\mathbb{R}$, this is true:

$$1, \quad 1+1, \quad 1+1+1, \quad \ldots$$

never returns to 0. However, in finite fields such as $\mathbb{F}_5$, it is not true:

$$1+1+1+1+1=0.$$

Thus, under the map

$$\mathbb{Z} \to \mathbb{F}_5,$$

the integer 5 maps to 0. In fact,

$$\ldots, -10, -5, 0, 5, 10, \ldots$$

all have the same image.

The **image** of this map is the set of elements of K that can actually be attained from applying ϕ to the integers.

Definition 1.1.7 (Image) — Let A, B be sets and define function $\phi : A \to B$. We define the *image* of ϕ as

$$\text{im}(\phi) = \{\phi(a) : a \in A\}.$$

It is always a subset of B, the codomain.

So, the set of elements of K that can be attained by ϕ is

$$\text{im}(\phi) = \{\phi(n) : n \in \mathbb{Z}\}.$$

Example 1.1.8. If $K = \mathbb{Q}$, then the image of ϕ is just the ordinary integers taken inside $\mathbb{Q}$.

If $K = \mathbb{F}_5$, however, the image is

$$\{0, 1, 2, 3, 4\},$$

because anything outside this set maps modulo 5 to something inside it.

Serre says that this image is an **integral domain**.

Definition 1.1.9 (Integral Domain) — An **integral domain** is a commutative ring in which the product of two nonzero elements is never 0. Equivalently,

$$ab = 0 \implies a = 0 \text{ or } b = 0.$$

This property is often described by saying that the ring has *no zero divisors*. For instance, in $\mathbb{Z}/6\mathbb{Z}$, we saw that $2 \cdot 3 = 0$. However, $2, 3 \neq 0$, so 2 and 3 are zero divisors.

Example 1.1.10. $\mathbb{Z}$ is an integral domain. If two integers are both nonzero, their product cannot be 0. On the other hand, $\mathbb{Z}/6\mathbb{Z}$ is not an integral domain, because

$$2 \cdot 3 \equiv 0 \pmod 6,$$

but $2, 3 \not\equiv 0 \pmod 6$.

Problem 1.1.11

Show that all fields are integral domains.

Solution. *Let K be a field, and let $a, b \in K$. Assume that $a \neq 0$. Then there exists a^{-1} in K such that $aa^{-1} = 1$. Assume that*

$$ab = 0.$$

Then

$$a^{-1}(ab) = a^{-1} \cdot 0.$$

We can show from the ring axioms that $a^{-1} \cdot 0 = 0$. By associativity,

$$a^{-1}(ab) = (a^{-1}a)(b),$$

and by commutativity,

$$(a^{-1}a)(b) = (aa^{-1})(b) = (1)(b) = (b)(1) = b.$$

Hence $b = 0$, so K is an integral domain.

⋆ **Problem 1.1.12** (Bonus)

Prove that every finite integral domain R is a field.

Hint. For a nonzero $a \in R$, consider the function

$$\phi : R \to R, \qquad \phi(x) = ax.$$

Which steps of your proof do/do not generalize to infinite integral domains? Give an example of an infinite integral domain that is not a field.

Why must the image of $\mathbb{Z}$ in a field K be an integral domain?

First, we establish that $\mathrm{im}(\phi)$ is a commutative ring. Verifying the ring axioms, let

$$a = \phi(m), \qquad b = \phi(n),$$

then

$$a + b = \phi(m + n), \qquad ab = \phi(mn), \qquad -a = \phi(-m).$$

Moreover, $0_K = \phi(0)$ and $1_K = \phi(1)$.

Because multiplication in $\mathrm{im}(\phi)$ is inherited from the field K, the image has no zero divisors, so $\mathrm{im}(\phi)$ is an integral domain.

This motivates the following definition.

Definition 1.1.13 (Characteristic) — Let K be a field. If there is a positive integer n such that

$$n \cdot 1_K = 0,$$

then the smallest such positive integer is called the **characteristic** of K.

If no such positive integer exists, we say that K has characteristic 0.

Problem 1.1.14

Find $\mathrm{char}(\mathbb{Q}), \mathrm{char}(\mathbb{R})$, and $\mathrm{char}(\mathbb{F}_7)$.

Solution.

$$\mathrm{char}(\mathbb{Q}) = \mathrm{char}(\mathbb{R}) = 0, \qquad \mathrm{char}(\mathbb{F}_7) = 7.$$

Current Progress

The current draft ends here. In the next portion, we will show that every finite field's characteristic is a prime number p. This will allow us to identify the image of $\mathbb{Z}$ in K with either $\mathbb{Z}$ or $\mathbb{F}_p$, and will lead to discussions of isomorphisms, homomorphisms, and Serre's first lemma concerning the Frobenius map

$$x \longmapsto x^p.$$

AI Use Statement

In writing this guide, generative AI tools were used for researching and editing. Their uses included assisting with background research and terminology, brainstorming practice problems and examples, checking computations and logic, and providing feedback on wording and organization. I independently verified all mathematical content.

The mathematical perspective and pedagogical approach of this guide are my own, including the choices of what to emphasize, the connections and clarifying/admonitory remarks, choice of practice problems, etc.